

Important Note: Please answer all the questions in English and submit a photocopy of your answer to Midterm at learn.tsinghua.edu.cn before 9:50 AM, April 15 2020. Although you can use books or online resources, you are **NOT ALLOWED** to discuss with your classmate or resort to other's help by any means about the problems.

Problem 1. (10 points) (Discrete random variables)

Please judge the correctness of the following proposition and prove your conclusion. Use the *definition* of expectation in your derivation.

- a) $\mathbb{E}(Y|X) \geq 0$ if $Y \geq 0$
- b) $\mathbb{E}(aY + bZ|X) = a\mathbb{E}(Y|X) + b\mathbb{E}(Z|X)$ for $a, b \in \mathbb{R}$
- c) $\mathbb{E}(Yg(X)|X) = g(X)\mathbb{E}(Y|X)$
- d) $\mathbb{E}\{\mathbb{E}(Y|X, Z)|X\} = \mathbb{E}(Y|X) = \mathbb{E}\{\mathbb{E}(Y|X)|X, Z\}$

Problem 2. (12 points) (Event and Probability)

Alice is gambling with some dices, while the number of dices is a random variable. The bonus dollar Alice wins is the sum of the outcomes of these dices. Let N be the number of the dices, B_i be the event that $N = i$, and C be the amount of dollars Alice wins. The probability of B_i is given by $\mathbb{P}(B_i) = 2^{-i}, i \geq 1$.

- a) What is the probability of $N = 2$ given that Alice wins 4 dollars?
- b) What is the probability of $C = 4$ given that the number of dices N is even?
- c) What is the probability of $N = 2$, given that $C = 4$ and the first outcome is 1?
- d) What is the probability that the largest outcome is r ?

Problem 3. (10 points) (Continuous random variables)

Many real-world problems are difficult to solve directly, and thus the problem of making a good estimation arises. Suppose the value of $\mathcal{L} = \int_0^1 h(x)dx$ is demanded, with $0 \leq h(x) \leq 1$ for all x . Suppose X and Y are independent random variables uniformly distributed on $[0, 1]$. Here we give three different derived variables: 1) $R = 1_{\{Y \leq h(X)\}}$ (indicator function), 2) $S = h(X)$, 3) $T = 1/2\{h(X) + h(1 - X)\}$.

- a) Calculate the expectation of the three random variables.
- b) Compare the variance of the three random variables. Which one do you think is the best estimator?

Problem 4. (12 points) (Discrete random variables)

The coronavirus, or COVID-19, has sickened more than a million people, forced millions to live under strict social distancing orders and paralyzed the global economy lately. Assume that you are visiting a city with 10^5 residents, among which any given individual may be infected by COVID-19 with probability 10^{-5} independently of all the others.

- a) What is the probability that there is at least one other infected person given that the authority finds one?
- b) What is the probability that there is at least one more if the authority finds two such people?
- c) Based on (a) and (b), how many infected people need to be found before the authority can be reasonably confident that they have found them all, i.e., with small probability r that there are no more infected people? Please first give the implicit relationship between the number of such people and r , and then provide a rough estimate for the number of such people for $r = 0.05$.

(**Hint:** Consider Binomial and Poisson distributions)

Problem 5. (12 points) (Continuous random variables)

Suppose we choose three points independently at random on the circumference of a circle. Denote by $A(x)$ the probability that at least one of the angles of this triangle is larger than $x\pi$.

- a) What is the value of $A(x)$ as $x \in [1/3, 1/2]$?
- b) What is the value of $A(x)$ as $x \in [1/2, 1]$?
- c) Calculate the density and expectation of the largest angle in the triangle.

Problem 6. (15 points) (Independence)

Ever since the emergence of coronavirus, referred to as COVID-19, it is recommended that people should not have friends and extended family over, or go to their homes. However, four reckless people, say Alex, Bob, Cauchy, David, are still hanging out at a club. Each of the 6 pairs, e.g. Alex and Bob, have **no discussion** with probability p and have a discussion with probability $1 - p$. Suppose the discussion take place independently of each pair. In future, if any of them get infected, he would have infected the person he talked to. If initially only Alex is infected by the virus, what is the probability that:

- a) David is infected given Bob and Cauchy haven't talked;
- b) David is infected given Alex and Bob haven't talked;

- c) David is infected by the virus.

Problem 7. (15 points) (Multiple discrete random variables)

Lily lives in a small island and needs to cross the strait by boat to reach the town, where her sister lives. The captain is superstitious about legends and only sets off when having the same weather for three consecutive days, either three consecutive sunny or rainy days. It's known that the weather is sunny with probability p and rainy with probability $1 - p$, independent everyday.

- a) What is the expected number of days Lily has to wait for her voyage?
b) Along with (a), what is the expected number of waiting days if $p = 1/3$?

(**Hint:** Consider the expected number of waiting days given the first day is sunny or raining.)

Problem 8. (10 points) (Continuous random variables)

Tom the cat is chasing Jerry the mouse all the way across an amusement park. When they reach a station, Jerry jumps onto a train which passes a nearby station every 8 minutes. Tom finds a bumper car but it may be required registration at the station with the probability r . If the registration isn't required, he continues his chasing. Otherwise he waits for registration to finish and continues his chasing with the bumper car, or he boards the train with Jerry circling back (8 minutes later) if it happens before the registration is over. The registration takes a time uniformly distributed between 0 and 15 minutes.

Let $r = \frac{n+1}{n+3}$ be a constant generated from the last digit of your student ID n . For example, if your ID is 2017018492, then $r = \frac{2+1}{2+3} = 3/5$

- a) What is the probability that Tom aboard the train with Jerry circling back.
b) Derive the CDF of Tom's waiting time.
c) Calculate the expected value of Tom's waiting time.